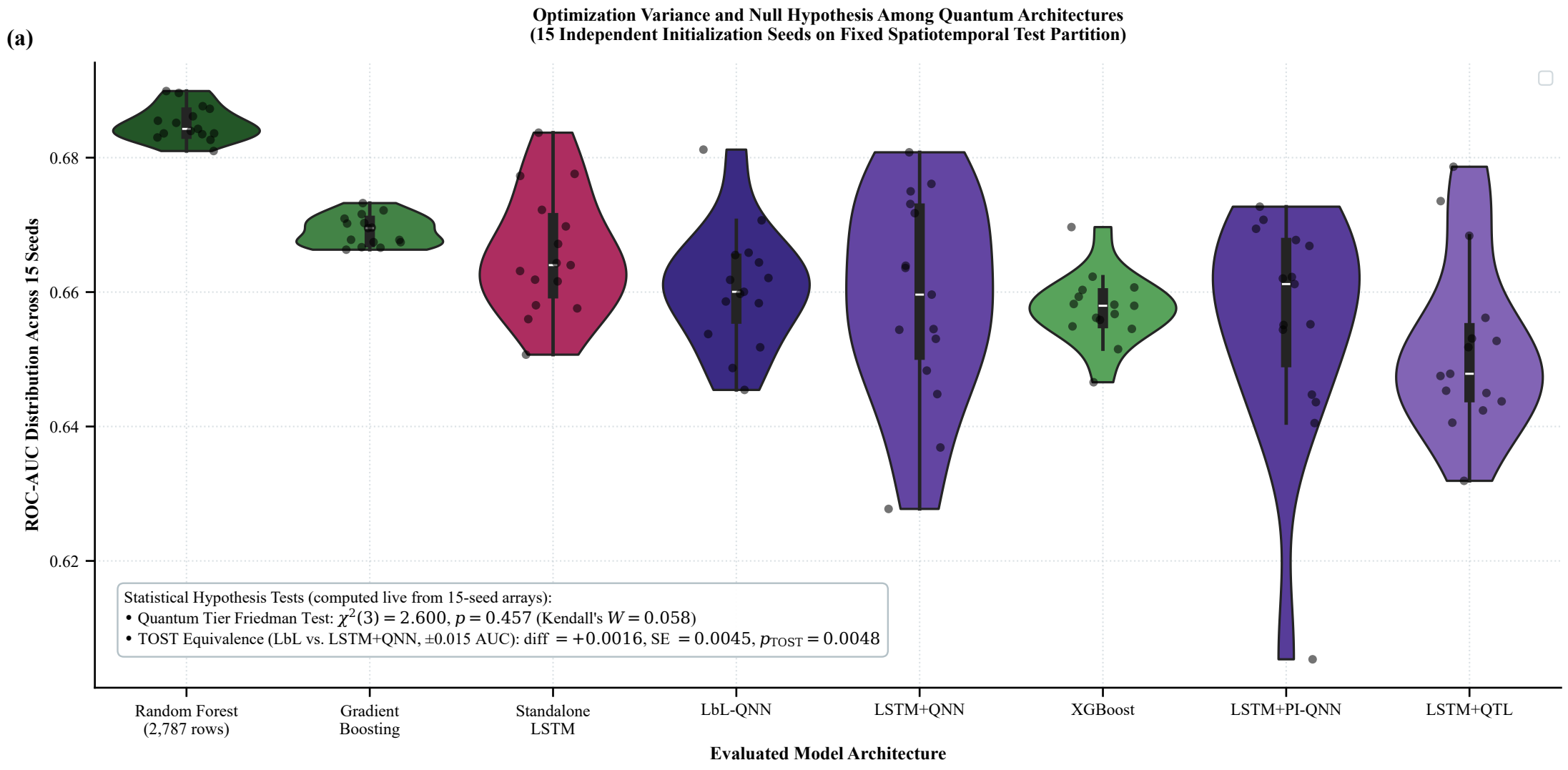

(a)
Optimization Variance and Null Hypothesis Among Quantum Architectures
(15 Independent Initialization Seeds on Fixed Spatiotemporal Test Partition)
ROC-AUC Distribution Across 15 Seeds
0.68
0.66
0.64
0.62
Statistical Hypothesis Tests (computed live from 15-seed arrays):
• Quantum Tier Friedman Test: $\chi^2(3) = 2.600$, $p = 0.457$ (Kendall's $W = 0.058$)
• TOST Equivalence (LbL vs. LSTM+QNN, ±0.015 AUC): diff $= +0.0016$, SE $= 0.0045$, $p_{\text{TOST}} = 0.0048$
Random Forest (2,787 rows)
Gradient Boosting
Standalone LSTM
LbL-QNN
LSTM+QNN
XGBoost
LSTM+PI-QNN
LSTM+QTL
Evaluated Model Architecture